

HIMAN WITHANA

AI/ML Engineer | Applied AI & Full Stack AI Engineering

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PROFESSIONAL SUMMARY

AI/ML Engineer with over a year of full-time industry experience designing, building, and deploying production AI systems for enterprise clients. Specializes in applied machine learning, generative AI and large language model applications, and computer vision, combined with the full-stack engineering ability to take a model from experimentation through to a live, revenue-generating product. Owns delivery end to end – problem framing, model selection, backend services, third-party integrations, and user-facing interfaces – and translates business requirements into reliable, scalable solutions. Works independently on ambiguous technical problems and delivers to production deadlines in fast-moving commercial environments.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, SQL, Dart, R, HTML/CSS

Machine Learning & Computer Vision: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, YOLOv8/v11 (detection & segmentation), U-Net, EfficientNet, ResNet, MobileNet, Segment Anything Model (SAM), SAHI, transfer learning, knowledge distillation, model benchmarking, data annotation (CVAT, Supervisely, Label Studio)

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, OpenAI GPT-4o / GPT-4o-mini, Google Gemini, Hugging Face, prompt engineering, embeddings, semantic search, hallucination mitigation, Model Context Protocol (MCP), agentic workflows

Backend & Data: FastAPI, Flask, REST APIs, WebSocket, Server-Sent Events (SSE), microservices, async Python, OAuth 2.0, webhooks; PostgreSQL, pgvector, Milvus, FAISS, Firebase, pandas, NumPy, ETL, Apache Airflow, Spark, Kafka

Frontend, Cloud & DevOps: React, Nuxt 3, Vue 3, Vuetify, Tailwind CSS, Flutter, embeddable JS widgets; AWS (EC2, S3, IAM), Docker, CUDA/GPU acceleration, Linux, Git, Jira, Agile

PROFESSIONAL EXPERIENCE

Junior AI/ML Engineer (Full-Time)

February 2026 – August 2026

Avya Technologies (Pvt) Ltd, Colombo, Sri Lanka

Enterprise Multi-Tenant RAG Chatbot Platform (SaaS)

(Python, FastAPI, LangChain, GPT-4o-mini, Milvus, PostgreSQL, Docker, AWS, Nuxt 3)

- Architected and built the microservice backend behind a multi-tenant enterprise RAG chatbot platform now serving live customers in production on AWS.
- Designed the document ingestion and knowledge-base pipeline that allows clients to onboard their own content and keep it accurate as their documents change.
- Improved answer accuracy and reduced hallucinated responses by redesigning the retrieval and prompting layer, and cut perceived response times by moving the product to real-time streaming replies.
- Delivered the commercial layer end to end – subscription plans, usage entitlements, payment integration, and admin dashboards – enabling the platform's first paying customers.
- Built the customer-facing web application and the embeddable chat widget deployed on client websites, owning the work from API design through to production UI.

Model Context Protocol (MCP) Server – Tool-Calling Agent

(Python, MCP, PostgreSQL, REST APIs, LLM tool calling, prompt engineering)

- Led development of an MCP server connecting the chatbot to live enterprise databases and APIs, extending the product from static document search to answering real-time operational questions.

- Integrated the agent into the existing conversation flow without disrupting production behavior, and tuned tool definitions and prompting to make tool calling reliable enough for client use.

Omnichannel Messaging Integration – WhatsApp Business & Meta Messenger

(Python, FastAPI, Meta Graph API, webhooks, OAuth 2.0, async processing)

- Expanded the platform to WhatsApp Business and Facebook Messenger, giving clients AI-powered support on the channels their customers already use.
- Owned Meta platform compliance – self-service page connection, data-deletion handling, and policy requirements – securing Business Verification and unblocking the channel launch.

Real-Time OCR Attendance System

(Python, FastAPI, Google Gemini Vision OCR, WebSocket, Nuxt/Vue)

- Delivered a full-stack OCR attendance solution for an enterprise client, combining mobile capture, vision-based document extraction, and real-time delivery of results to the web application.

Junior AI/ML Engineer, R&D (Full-Time)

July 2025 – January 2026

Turbo Gen Consultancy (Pvt) Ltd, Colombo, Sri Lanka

Computer Vision Instance Segmentation for Agricultural Imaging

(PyTorch, TensorFlow, YOLOv8/v11-seg, U-Net, SAM, SAHI, OpenCV, FastAPI, Flutter)

- Delivered an end-to-end instance segmentation system for agricultural crop imaging, evaluating multiple detection and segmentation architectures to find the best accuracy-to-performance balance for on-device deployment.
- Improved model accuracy on small and overlapping objects through targeted data preparation, augmentation design, and advanced fine-tuning techniques.
- Deployed the model as a production inference service and integrated it into a mobile application, including model conversion for edge devices.

Automated Data Annotation Pipeline

(Python, YOLOv8-seg, SAM, CVAT, Supervisely, Docker, CUDA)

- Cut manual labeling effort substantially by building an automated annotation pipeline that used the trained model to pre-label new images for human review, and set up the team's annotation and GPU infrastructure to reduce cloud compute costs.

RESEARCH EXPERIENCE

Individual Research Project

2026 – Present

Uncertainty Calibration in Bayesian Optimization | Robert Gordon University / Informatics Institute of Technology

- Investigating how uncertainty calibration in probabilistic models affects the efficiency of automated optimization systems, and which correction methods best restore lost performance.
- Designed and implemented the full experimental framework in Python, including the benchmarking methodology and evaluation against current published approaches.

SELECTED PROJECTS

Agentic Data Extraction System *(Python, LangChain, LangGraph, Playwright)*

- Built an autonomous LLM agent that turns natural-language instructions into multi-step web automation plans, with structured output, automated report generation, and recovery from failures.

AI-Powered Waste Management System *(Python, BiLSTM, LangChain, FAISS, Flask)*

- Built an AI-driven waste collection platform combining natural language understanding, conversational query answering, demand forecasting, and route optimization.

Automated Weather Data ETL Pipeline *(AWS EC2, S3, Apache Airflow, Python)*

- Built an automated cloud data pipeline that collects, processes, and stores live weather data on a scheduled workflow.

EDUCATION

BSc (Hons) Artificial Intelligence and Data Science

2023 – 2027 (Expected)

Robert Gordon University, Aberdeen, UK – delivered at Informatics Institute of Technology, Colombo, Sri Lanka

- **Relevant coursework:** Machine Learning, Deep Learning, NLP, Computer Vision, Edge AI, Data Engineering, Advanced Mathematics for Data Science, Data Structures and Algorithms, Database Systems.

G.C.E. Advanced Level Examination

2020 – 2022

- Successfully completed the Physical Science stream: Combined Mathematics, Physics, Chemistry, and English.

G.C.E. Ordinary Level Examination

2014 – 2019

- Successfully completed in the English medium, including Mathematics and English.

AWARDS & RECOGNITION

- **1st Place – Cutting Edge 2025 (VisionQuest)**, Data Science Group Project category, Informatics Institute of Technology.
- **Nominee – National ICT Awards (NBQSA) 2025**, British Computer Society Sri Lanka.
- **Hackathons:** IEEEExtreme 18.0 (IEEE Region 10), Virtusa Capture the Flag, CodeSprint 8 Ideathon (IEEE SB IIT).

LEADERSHIP & COMMUNITY

- **Coordinator, AI-Driven Sri Lanka – IEEE Young Professionals Sri Lanka (2025 – 2026):** Planned and ran national AI education workshops, managing technical programming and volunteer logistics.
- **Developer, IEEE Computational Intelligence Society – IIT Student Chapter (2025 – 2026):** Led development of web-based and AI-driven projects and technical resources for the chapter.

REFERENCES

Mr. Pasindu Madhuranga

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